

Gears

For the latest information about the unit price, lead time and catalog standard, please visit our Web site.

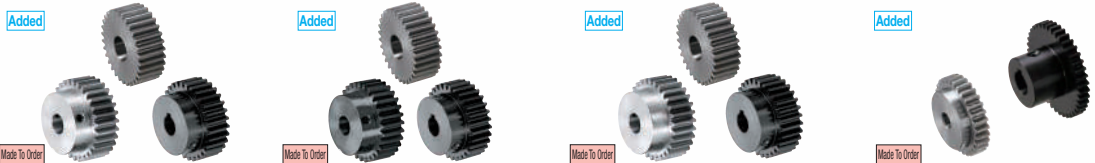
Spur Gears



Product Name	Induction Hardened, Ground - Module 1.0, 1.5, 2.0, 2.5, 3.0	Shape Shaft Bore Fixed - Module 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0	Shaft Bore Configurable - Module 0.5
Page	1535	1537	1539



Product Name	Shaft Bore Configurable - Module 0.8	Shaft Bore Configurable - Module 1.0, Standard	Shaft Bore Configurable - Module 1.0, Thin	Shaft Bore Configurable - Module 1.5
Page	1540	1541	1542	1543



Product Name	Shaft Bore Configurable - Module 2.0	Shaft Bore Configurable - Module 2.5	Shaft Bore Configurable - Module 3.0	- Tooth Width, Hub Dimensions Configurable
Page	1545	1547	1549	1551



Product Name	Plastic - Module 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0	Bonded Plastic - Module 1.0, 1.5, 2.0, 2.5, 3.0	Keyless Hub Spur Gears - Module 1.0, 1.5, 2.0, 2.5, 3.0 (Steel)	Bearing Built-In - Module 1.0, 1.5, 2.0
Page	1553	1555	1557	1559

Rack Gears



Product Name	Induction Hardened Rack Gears - Ground	Induction Hardened Rack Gears - Ground, Hole Position Configurable	Rack Gears - L Fixed
Page	1561	1562	1563



Product Name	Rack Gears - L Configurable (One End Machined)	Round Rack Gears - L Fixed/Configurable	Product Name	Worm	Worm Wheel
Page	1564	1565	Page	1566	1566



Product Name	Bevel Gears - Module 1.0, 1.5, 2.0	Helical Gears - Module 1.0, 1.5, 2.0, 2.5, 3.0	Non-Contact Magnetic Transmission Drives Standard/Economy Type
Page	1567	1568	1569-1570

Gears

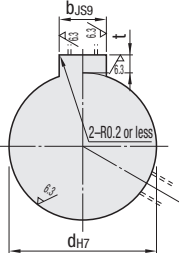
Overview / Technical Data, continued

Allowable Transmission Power (Bending Strength) Calculation

Material	1045 Carbon Steel or Equivalent	1045 Carbon Steel or Equivalent	304 Stainless Steel	Free-Cutting Brass Bar	MC Nylon	Polyacetal
	—	Tooth Surface Induc. Hardened				
Formula	JGMA401-01				Lewis Formula	
Mating Gear	Same Material, Same No. of Teeth				—	Metal Material
Speed	100 rpm	500 rpm	100 rpm		100 rpm	
Lubrication Type	—	—	—	—	Non-Lubricated	
Ambient Temperature	—	—	—	—	40°C	20°C
Stress Cycles	More than 107 times				—	107
Impact from Motor	Equal Load					Equal Load
Impact from the Mating Gear	Equal Load					Equal Load
Load Direction	Bidirectional					—
Allowable Tooth Root Bending Stress (kgf/mm²) *	18.4	23.0	10.5	4.0		
Safety Factor	1.2					

* Allowable Tooth Root Bending Stress is 2/3 of a maximum fatigue limit because load is bidirectional.

Keyway Dimensions N: New JIS (B1301) Keyway Dimensions



Number	dH7	bJS9	t Tolerance
8N	8	3	±0.0125
10N	10	3	±0.0125
10K	10	4	1.8
11N	11	4	1.8
12N	12	4	1.8
13N	13	4	1.8
14N	14	5	2.3
15N	15	5	2.3
16N	16	5	±0.0150
17N	17	5	±0.0150
18N	18	5	±0.0150
19N	19	6	2.8
20N	20	6	2.8
21N	21	6	2.8
22N	22	6	2.8

Number	dH7	bJS9	t Tolerance
23N	23	8	±0.0210
24N	24	8	±0.0210
25N	25	8	±0.0210
26N	26	8	±0.0210
27N	27	8	±0.0210
28N	28	8	±0.0210
29N	29	8	±0.0210
30N	30	8	±0.0210
31N	31	10	±0.0180
32N	32	10	±0.0180
33N	33	10	±0.0180
34N	34	10	±0.0250
35N	35	10	±0.0250
36N	36	10	±0.0250
37N	37	10	±0.0250
38N	38	10	±0.0250

Number	dH7	bJS9	t Tolerance
39N	39	12	±0.0215
40N	40	12	±0.0215
41N	41	12	±0.0215
42N	42	12	±0.0215
43N	43	12	±0.0215
44N	44	12	±0.0215
45N	45	12	±0.0215
46N	46	12	±0.0215
47N	47	14	±0.0215
48N	48	14	±0.0215
49N	49	14	±0.0215
50N	50	14	±0.0215

Spur Gear

Pressure Angle 20°, Module 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0, B Shape, Shaft Bore Fixed Type

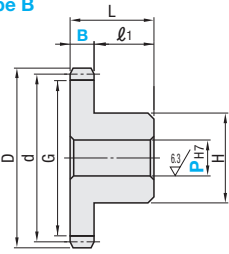
■ **Features:** Compared with Straight Bore (GEAHBB), price is reduced by maximum of 76% and by making it in stock, next-day shipment can be realized.



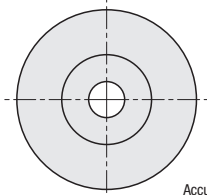
RoHS 10

Type	Material	Surface Treatment
GEAHF	S45C Equivalent	Black Oxide

Shape B



Shaft Bore Specifications



Accuracy Previous JIS B 1702 Class 4
(New JIS B 1702-1 Class 8 Equivalent)

■ **Module 0.5/0.8/1.0** (For Shaft Bore Configurable Types, see **P.1539~1542**. For Tooth Width, Hub Dimensions Configurable Type, see **P.1551**.)

Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. PH7	d	D	G	H	L	ℓ1	Unit Price	
Type	Module					Reference Dia.	Tip Dia.	Root Dia.					
GEAHF	0.5	28	3	B	3	14	15	12.75	10	8	5		
		30				15	16	13.75					
		32				16	17	14.75					
		35				17.5	18.5	16.25					
		36				18	19	16.75					
		40	2		3	20	21	18.75	10	7	5		
		42				21	22	19.75					
		45				22.5	23.5	21.25					
		48				24	25	22.75					
		50				25	26	23.75					
		52	2		5	26	27	24.75	20	7	5		
		60				30	31	28.75					
		70				35	36	33.75					
		80				40	41	38.75					
		100				50	51	48.75					
	0.8	20	7		4	16	17.6	14	10	14	7		
		24	7		5	19.2	20.8	17.2	12.5	14	7		
		25				20	21.6	18					
		28				22.4	24	20.4					
		30				24	25.6	22					
		32	5		5	25.6	27.2	23.6	12.5	14	9		
		36	5		6	28.8	30.4	26.8	14	14	9		
		40				32	33.6	30					
		45				36	37.6	34					
		48				38.4	40	36.4					
		50				40	41.6	38					
	1.0	20	12		6	20	22	17.5	16	20	8		
		21	12		6	21	23	18.5	18	20	8		
		22	12		6	22	24	19.5	20	20	8		
		23				23	25	20.5					
		24				24	26	21.5					
		25				25	27	22.5					
		26	12		6	26	28	23.5	22	20	8		
		27	12		6	27	29	24.5	24	20	8		
		28	12		8	28	30	25.5	24	20	8		
		29	12		6	29	31	26.5	26	20	8		
		30	12		8	30	32	27.5	27	20	8		
		32	10		8	32	34	29.5	28	20	10		
		34				34	36	31.5					
		35				35	37	32.5					
		36				36	38	33.5					
		38				38	40	35.5					
		40	10		8	40	42	37.5	30	20	10		
		42				42	44	39.5					
		44				44	46	41.5					
		45				45	47	42.5					
		46				46	48	43.5					
		48	10		8	48	50	45.5	44	20	10		
		50	10		8	50	52	47.5	46	20	10		
		52				52	54	49.5					
		54				54	56	51.5					
		55				55	57	52.5					
		56				56	58	53.5					
		58	10		10	58	60	55.5	50	20	10		
		60				60	62	57.5					
		62				62	64	59.5					
		64				64	66	61.5					
		65				65	67	62.5					
		66	10		10	66	68	63.5	56	20	10		
		68				68	70	65.5					
		70				70	72	67.5					
		72				72	74	69.5					
		75				75	77	72.5					
		80	10		10	80	82	77.5	60	20	10		
		84				84	86	81.5					
		85				85	87	82.5					
		90				90	92	87.5					
		95				95	97	92.5					
		96	10		16	96	98	93.5	70	20	10		
		100				100	102	97.5					
		110				110	112	107.5					
		120				120	122	117.5					

Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837	P1-901	P1-1521	P2-277	P1-947	P1-1017	P1-297
P1-900	P1-926	P1-1530	P2-282	P1-1014	P1-1049	P1-330

■ **Module 1.5/2.0/2.5/3.0** (For Shaft Bore Configurable Types, see **P1543~1549**. For Tooth Width, Hub Dimensions Configurable Type, see **P1551**.)

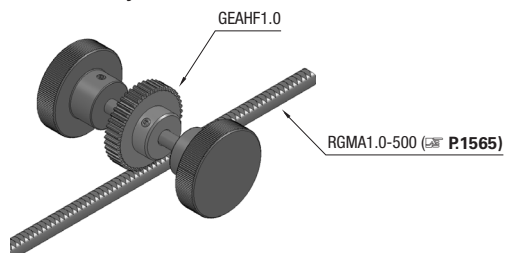
Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. PH7	d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ1	Unit Price
Type	Module											
GEAHF	1.5	20	15	B	6	30	33	26.25	25	27	12	
		21				31.5	34.5	27.75				
		22				33	36	29.25	26			
		23				34.5	37.5	30.75	27			
		24				36	39	32.25	30			
		25				37.5	40.5	33.75	32			
		26				39	42	35.25	34			
		27				40.5	43.5	36.75	35			
		28				42	45	38.25				
		29				43.5	46.5	39.75				
		30				45	48	41.25				
		32				48	51	44.25				
		34				51	54	47.25				
		35				52.5	55.5	48.75	40			
		36				54	57	50.25				
		38				57	60	53.25				
		40				60	63	56.25				
		42			63	66	59.25	45				
		44			66	69	62.25					
		45			67.5	70.5	63.75					
		46			69	72	65.25					
		48			72	75	68.25					
		50			75	78	71.25					
		52			78	81	74.25	50				
		54			81	84	77.25					
		55			82.5	85.5	78.75					
		56			84	87	80.25					
		58			87	90	83.25					
		60			90	93	86.25					
		62			93	96	89.25					
		64			96	99	92.25	55				
		65			97.5	100.5	93.75					
		66			99	102	95.25					
	2.0	21	20		8	42	46	37	34	34	14	
		22				44	48	39	36			
		23				46	50	41	37			
		24				48	52	43	40			
		25				50	54	45	42			
		26				52	56	47	42			
		27				54	58	49	45			
		28				56	60	51	45			
		29				58	62	53	47			
		30				60	64	55	48			
		32			64	68	59	50				
		34			68	72	63	50				
		35			70	74	65	52				
		36			72	76	67	55				
		38			76	80	71					
		40			80	84	75					
		42			84	88	79	60				
		44			88	92	83					
		45			90	94	85					
		46			92	96	87					
	2.5	20	25		8	50	55	43.75	40	41	16	
		21				52.5	57.5	46.25	42			
		22				55	60	48.75	44			
		23				57.5	62.5	51.25	46			
		24				60	65	53.75	48			
		25				62.5	67.5	56.25	50			
		26			65	70	58.75	55				
		27			67.5	72.5	61.25	60				
		28			70	75	63.75					
		29			72.5	77.5	66.25	62				
		30			75	80	68.75	65				
		32			80	85	73.75					
	34	85	90		78.75	70						
	35	87.5	92.5		81.25							
	36	90	95		83.75	75						
	3.0	20	30		8	60	66	52.5	50	48	18	
		21				63	69	55.5	52			
		22				66	72	58.5	54			
		23				69	75	61.5	56			
		24				72	78	64.5	58			
		25			75	81	67.5	60				
		26			78	84	70.5					
		27			81	87	73.5	65				
28		84		90	76.5							
29		87		93	79.5	70						
30		90		96	82.5	75						



Ordering
Example

Part Number - Number of Teeth - B - Gear Shape - Shaft Bore Dia.
GEAHF1.0 - 40 - 10 - B - 8

Pinion mechanism by Round Gear Rack

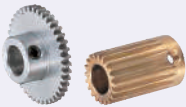


Spur Gear

Pressure Angle 20°, Module 0.5, Shaft Bore Configurable Type

Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837	P1-901	P1-1521	P2-277	P1-947	P1-1017	P1-297
P1-900	P1-926	P1-1530	P2-282	P1-1014	P1-1049	P1-330

For products uncovered by the e-Catalog Standard, see P131.



Type	Material	Surface Treatment	Accessory
Straight Bore	S45C Equivalent	Black Oxide	Set Screw (SCM435 Black Oxide)
Straight Bore + Tap	Free-Cutting Brass Bar SUS304	Electroless Nickel Plating	Set Screw (SUS304)

Set Screw is not included in Un-tapped Type products.

Shaft Bore Specifications (Selectable Gear Shapes)

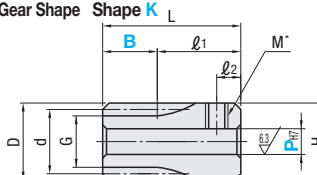
Straight Bore (Shape A, Shape B, Shape K)



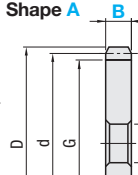
Straight Bore + Tap (Shape B, Shape K)



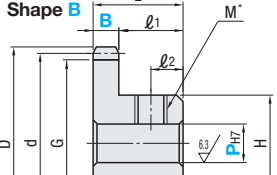
Gear Shape Shape K



Shape A



Shape B



Accuracy Previous JIS B 1702 Class 4 (New JIS B 1702-1 Class 8 Equivalent)

⊗ Tapped shaft bores are not available for Shape A.

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on P1557. For configuring the tooth width / hub dimensions, see P1551.

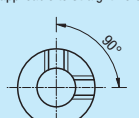
Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. Per (1mm Increment)	d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ ₁	ℓ ₂	M (Coarse)	*1 Allowable Transmission Force (N · m) Bending Strength			Unit Price						
Type	Module													Straight Bore			Straight Bore + Tap						
															S45C Equivalent	Free-Cutting Brass Bar	SUS304	GEAHB	GEABN	GEABB	GEABG	GEAB	GEABS
(Selectable Gear Shapes) Straight Bore (Shape A, Shape B, Shape K) GEAHB Straight Bore + Tap (Shape B, Shape K) GEABN GEABB GEABG GEAB GEABS	0.5	15	8	K	3~5	7.5	8.5	6.25	9	18	10	3	M3	0.72	0.16	0.41							
		16				8	9	6.75						0.79	0.17	0.45							
		18				9	10	7.75						10	0.95	0.21	0.54						
		20				10	11	8.75						11	1.12	0.24	0.64						
		20	3	A	3~6, 6.35	10	11	8.75	8.5	8	10	5		2.5	0.42	0.09	0.24						
		24				12	13	10.75							0.54	0.12	0.31						
		25				12.5	13.5	11.25							0.58	0.13	0.33						
		26				13	14	11.75							0.61	0.13	0.35						
		28				14	15	12.75							0.68	0.15	0.39						
		30				15	16	13.75							0.74	0.16	0.42						
		32				16	17	14.75							0.8	0.17	0.46						
		35				17.5	18.5	16.25							0.91	0.2	0.52						
		36				18	19	16.75							0.94	0.2	0.54						
		40				20	21	18.75							0.72	0.16	0.41						
		42				21	22	19.75							0.76	0.17	0.43						
		45				22.5	23.5	21.25							0.83	0.18	0.48						
		48				24	25	22.75							0.9	0.2	0.51						
		50				25	26	23.75							0.95	0.21	0.54						
		*52	2	B	5~16	26	27	24.75	20	7				0.99	0.22								
		*60				30	31	28.75						1.18	0.26								
		*70				35	36	33.75						1.42	0.31								
		*80				40	41	38.75						1.65	0.36								
		*100				50	51	48.75						2.13	0.46								
		*120				60	61	58.75						2.59	0.56								

* marked number of teeth is not available for GEABS. ⚠ Shaft Bore Dia. 6.35 is available. ⚠ The "-" text on the above table means that any Shaft Bore Dia. is not selectable.

*1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions. For conditions, see P1534.

Ordering Example	Part Number	Number of Teeth	B	Gear Shape	P
	GEAB0.5	20	3	B	3
	GEAHB0.5	30	3	A	6
	GEABS0.5	16	8	K	5

Alterations	Part Number	Number of Teeth	B	Gear Shape	P	(KC90, KTC--etc.)
	GEAB0.5	20	3	B	3	KC120
	GEAHB0.5	50	2	A	5	KC15-K3.0

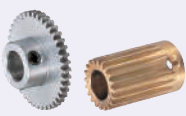
Alterations	Set Screw		Side Through Hole	Tapped Hole Dimension
Code	KC90	KC120	KFC, KTC	TPC
Spec.	Adds another set screw at 90° position. ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type. 	Adds another set screw at 120° position. ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type. 	Machines through holes on the side surface. (KFC, KTC: 1mm Increment, K: 0.5mm Increment) ⊗ Applicable to Shape A only. ⊗ $P+K+4 \leq KFC(KTC) \leq G-K-4$ [K Selection] K3.0~K6.0 [Ordering Code] KFC15-K3.5 	Changes the tapped hole dimension to M4. [Ordering Code] TPC4 ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type.

Spur Gear

Pressure Angle 20°, Module 0.8, Shaft Bore Configurable Type

Fixing Parts				Others			
Rotary Shaft	Cartleiver Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar	
P.1-837	P.1-901	P.1-1521	P.2-277	P.1-947	P.1-1017	P.1-297	
P.1-900	P.1-926	P.1-1530	P.2-282	P.1-1014	P.1-1049	P.1-330	

For products uncovered by the e-Catalog Standard, see P.131.



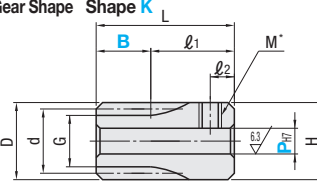
Type	Material	Surface Treatment	Accessory
Straight Bore	S45C Equivalent	Black Oxide	Set Screw (SCM435 Black Oxide)
Straight Bore + Tap	Free-Cutting Brass Bar SUS304	Electroless Nickel Plating	Set Screw (SUS304)

Shaft Bore Specifications (Selectable Gear Shapes)

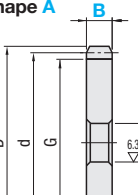
Straight Bore (Shape A, Shape B, Shape K)	Straight Bore + Tap (Shape B, Shape K)
	

Set Screw is not included in Un-tapped Type products.

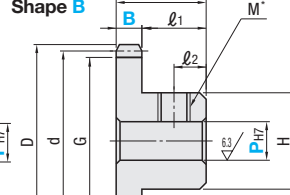
Gear Shape Shape K



Shape A



Shape B



Accuracy Previous JIS B 1702 Class 4 (New JIS B 1702-1 Class 3 Equivalent)

Tapped shaft bores are not available for Shape A.

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on P.1557. For configuring the tooth width / hub dimensions, see P.1551.

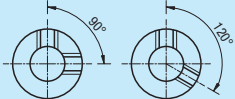
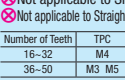
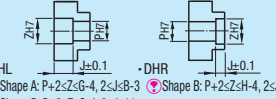
Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. P (mm Increment)	d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ1	ℓ2	M (Coarse)	*1 Allowable Transmission Force (N·m) Bending Strength	Unit Price								
Type	Module														Straight Bore + Tap						Straight Bore + Tap		
														S45C Equivalent	Free-Cutting Brass Bar	SUS304	GEAHB	GEABN	GEABB	GEABG	GEAB	GEABS	
(Selectable Gear Shapes)	Straight Bore (Shape A, Shape B, Shape K)	12	7	K	4, 5	9.6	11.2	7.6	11.2	20	13			M3	1.11	0.24	0.63						
		14				11.2	12.8	9.2	12.8						1.43	0.31	0.82						
		15				12	13.6	10	13.6						1.6	0.35	0.92						
		16				12.8	14.4	10.8	1.27						0.28	0.72							
		16				7	10	9	1.78						0.39	1.01							
		18				14.4	16	12.4	1.52						0.33	0.87							
		18				7	16	17.6	14						2.13	0.46	1.22						
		20				19.2	20.8	17.2	1.78						0.39	1.02							
		20				7	20	21.6	18						2.5	0.54	1.42						
		24				22.4	24	20.4	2.32						0.5	1.32							
	Straight Bore + Tap (Shape B, Shape K)	24	5	A	5-8	19.2	20.8	17.2	7	3.25	0.71	1.85											
		25	20			21.6	18	9	2.46	0.53	1.4												
		25	7			22.4	24	20.4	7	3.44	0.75	1.96											
		28	24			25.6	22	9	2.89	0.63	1.65												
		28	7			28.8	30.4	26.8	7	4.04	0.88	2.3											
		30	24			25.6	22	9	3.16	0.69	1.8												
		30	7			25.6	27.2	23.6	7	4.42	0.96	2.52											
		32	28.8			30.4	26.8	7	3.42	0.74	1.95												
		36	32			33.6	30	9	4.01	0.87	2.29												
		40	36			37.6	34	4	4.58	1	2.61												
		45	38.4			40	36.4		5.33	1.16	3.04												
		48	40			41.6	38		5.77	1.26	3.3												
		50							6.07	1.32	3.46												

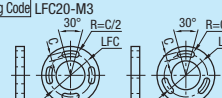
*1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions.

For conditions, see P.1534. Shaft Bore Dia. 6.35 is available. The "-" text on the above table means that any Shaft Bore Dia. is not selectable.

Ordering Example	Part Number	Number of Teeth	B	Gears Shape	P
	GEAB0.8	25	5	B	6
	GEABH0.8	30	7	A	8
	GEABS0.8	15	7	K	5

Alterations	Part Number	Number of Teeth	B	Gears Shape	P
	GEAB0.8	30	7	B	6
	GEABH0.8	40	5	A	8
					QTC16-M4

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore
Code	KC90, KC120	TPC	DHL, DHR	WDH
Spec.	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. Not applicable to Shape A. Not applicable to Straight Bore Type.	Changes the tapped hole dimension. Ordering Code TPC4 Not applicable to Shape A. Not applicable to Straight Bore Type.	Changes shaft bores to stepped bores. (Z: 1mm Increment, J: 0.1mm Increment) Ordering Code DHL-Z20-J2.0 Applicable to Straight Bore Type Only.	Changes shaft bores to both ends stepped hole. (Q, R, S, T: 1mm Increment) S, T: 3 Ordering Code WDH-Q10-R10-S5-T5
				

Alterations	Side Slotted Hole	Side Through Hole	Side Tapped Hole
Code	LFC, LTC	KFC, KTC	QFC, QTC
Spec.	Machines slotted holes on the side surface (30°). (LFC, LTC: 1mm Increment) P+C+4≤LFC(LTC)≤G-C-4 Applicable to Shape A only. M Selection M3, M4, M5, M6 Ordering Code LFC20-M3	Machines through holes on the side surface. (KFC, KTC: 1mm Increment) K: 0.5mm Increment Applicable to Shape A only. P+K+4≤KFC(KTC)≤G-K-4 K Selection K3.0-K6.0 Ordering Code KFC20-K3.5	Machines tapped holes on the side surface of the gear. (QFC, QTC: 1mm Increment) Applicable to Shape A only. P+M+4≤QFC(QTC)≤G-M-4 M Selection M3, M4 Ordering Code QFC25-M3 Tapped Hole Depth Mx1.5 (When B-Mx1.5, through)
			

Spur Gear

Pressure Angle 20°, Module 1.0, Shaft Bore Configurable Type

■ Standard Type and Thin Type are available.

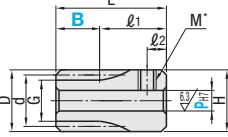


Type			Material	Surface Treatment	Accessory
Straight Bore	Straight Bore + Tap	Keyway, Keyway + Tap			
GEAHHB	GEAB	GEAKB	S45C Equivalent	-	Set Screw (SCM435, Black Oxide)
GEAHBB	GEABB	GEAKBB			
GEAHBG	GEABG	GEAKBG			
GEAHS	GEAS	GEAKS	SUS304	Electroless Nickel Plating	Set Screw (SUS304)

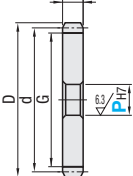
⚙️ Set Screw is not included in Un-tapped Type products.
 ⚙️ Selectable Gear Shapes differ depending on the number of teeth. Check the spec. table.

Gear Shape

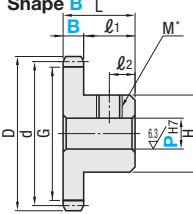
Shape K



Shape A



Shape B



Shaft Bore Specifications (Selectable Gear Shapes)

Straight Bore (Shape A, Shape B, Shape K)	Straight Bore + Tap (Shape B, Shape K)
Keyway (Shape A)	Keyway + Tap (Shape B)

⚙️ For Tap, Keyway Dimension Details, see **P1532**.
 ⚙️ Positioning of keyway and teeth is not fixed.

Accuracy Previous JIS B 1702 Class 4
(New JIS B 1702-1 Class 8 Equivalent)

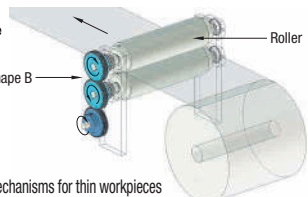
⚠️ Tapped shaft bores are not available for Shape A.

■ Standard Type

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on **P1557**. For configuring the tooth width / hub dimensions, see **P1551**.

Part Number	Number of Teeth	B	Gear Shape	Shaft Bore Dia. Ph7 (1mm Increment)		d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	l1	l2	M (mm)	*1 Allowable Transmission Force (N · m) Bending Strength		Unit Price											
				Straight Bore	Keyway, Keyway + Tap									S45C Equivalent	SUS304	Straight Bore			Straight Bore + Tap			Keyway Keyway + Tap					
Type	Module															GEAHHB	GEAHBB	GEAHBG	GEAHS	GEAB	GEABB	GEABG	GEAS	GEAKB	GEAKBB	GEAKBG	GEAKS

- ⚠️ * marked number of teeth is not available for stainless steel type (GEAHS, GEAS and GEAKS).
- ⚠️ Shaft bore diameter 9N is not available for Keyway Bore + Tap.
- ⚙️ Specify 10K as the P dimension if keyway width of 4.0mm (height 1.8mm) for Keyway + Tap with shaft bore diameter of 10 is desired **P1532**.
- ⚙️ Shaft bore diameter 6.35 is available for Straight Bore and Straight Bore + Tap. ⚙️ The "-" text on the above table means that any Shaft Bore Dia. is not selectable.
- *1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions. For conditions, see **P1534**. Calculation is performed based on 10mm of the tooth width.



As the Feed Mechanisms for thin workpieces

Fixing Parts					Others	
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837	P1-901	P1-1521	P2-277	P1-947	P1-1017	P1-297
P1-900	P1-926	P1-1530	P2-282	P1-1014	P1-1049	P1-330

For products uncovered by the e-Catalog Standard, see P.131.

Thin Type

Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. Ph7 (1mm Increment)		d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	L1	L2	M (Core)	1 Allowable Transmission Force (N·m) Bending Strength	Unit Price										
Type	Module				Straight Bore Straight Bore + Tap	Keyway, Keyway + Tap										S45C Equivalent	SUS304	Straight Bore		Straight Bore + Tap		Keyway Keyway + Tap				
Straight Bore (Shape A, Shape B, Shape K) GEAHB GEAHBB GEAHBG GEAHS	1	*14	6	K	6~10	-	14	16	11.5	16	25	19	17	4	M4	1.92	-									
		15					15	17	12.5	17		2.15				1.23										
		16					16	18	13.5	18		2.38				1.36										
		*17					17	19	14.5	19		2.62				-										
		18		A	6~12	18	20	15.5	20	2.86	1.63															
		19				19	21	16.5	16	3.09	1.77															
		20				20	22	17.5	18	3.34	1.91															
		21				21	23	18.5	20	3.59	2.05															
		22		8	6~12	22	24	19.5	20	3.85	2.19															
		*23				23	25	20.5	20	4.09	-															
24	24	26	21.5			22	4.35	2.48																		
25	25	27	22.5			22	4.61	2.63																		
26	A	6~15	8N-12N	26	28	23.5	22	4.87	2.78																	
*27				27	29	24.5	24	5.13	-																	
28				28	30	25.5	25	5.41	3.09																	
29				29	31	26.5	26	5.65	3.23																	
30	B	6~17	8N-15N	30	32	27.5	25	5.92	3.38																	
32				32	34	29.5	28	6.42	3.66																	
*34				34	36	31.5	25	7.01	-																	
35				35	37	32.5	28	7.28	4.15																	
36	6	8~17	8N-15N	36	38	33.5	25	7.51	4.29																	
*38				38	40	35.5	28	8.08	-																	
40				40	42	37.5	28	8.59	4.9																	
42				42	44	39.5	28	9.14	5.22																	
44	A	8~20	8N-17N	44	46	41.5	30	9.71	-																	
*45				45	47	42.5	30	10	-																	
46				46	48	43.5	30	10.29	-																	
48				48	50	45.5	30	10.83	6.18																	
50	B	8~28	10N-26N	50	52	47.5	40	11.37	6.49																	
*52				52	54	49.5	40	11.92	-																	
54				54	56	51.5	40	12.48	7.12																	
55				55	57	52.5	40	12.79	7.3																	
56	6	8~32	10N-30N	56	58	53.5	40	13.1	7.48																	
58				58	60	55.5	40	13.61	7.77																	
60				60	62	57.5	40	14.19	8.1																	
62				62	64	59.5	40	14.7	8.39																	
64	A	10~32	10N-30N	64	66	61.5	44	15.28	8.72																	
65				65	67	62.5	44	15.54	8.87																	
66				66	68	63.5	44	15.87	9.06																	
68				68	70	65.5	44	16.47	9.4																	
70	B	10~36	10N-34N	70	72	67.5	48	16.99	9.69																	
72				72	74	69.5	48	17.51	9.99																	
75				75	77	72.5	48	18.38	10.49																	
80				80	82	77.5	48	19.79	11.29																	
84	6	15~38	15N-36N	84	86	81.5	50	20.94	11.95																	
85				85	87	82.5	50	21.21	12.1																	
90				90	92	87.5	50	22.64	12.92																	
95				95	97	92.5	50	24.09	13.75																	
96	A	15~36N	15N-36N	96	98	93.5	54	24.36	13.9																	
100				100	102	97.5	54	25.56	14.58																	
110				110	112	107.5	54	28.27	16.13																	
120				120	122	117.5	54	31.12	17.76																	

* marked number of teeth is not available for stainless steel type (GEAHS, GEAS and GEAKS). Shaft bore diameter 9N is not available for Keyway Bore + Tap. The "-" text on the above table means that any Shaft Bore Dia. is not selectable.
Specify 10K as the P dimension if keyway width of 4.0mm (height 1.8mm) for Keyway + Tap with shaft bore diameter of 10 is desired. P.1532. Shaft bore diameter 6.35 is available for Straight Bore and Straight Bore + Tap.
*1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions. For conditions, see P.1534.
Calculation is performed based on 6mm of the tooth width.

Ordering Example	Part Number	Number of Teeth	B	Gear Shape	P
GEAB1.0	- 25	- 12	- B	- 8	
GEAKB1.0	- 38	- 10	- A	- 10N	
GEAS1.0	- 18	- 6	- K	- 10	

Alterations	Part Number	Number of Teeth	B	Gear Shape	P	(KC90, KC120, QFC, QTC-- etc.)
GEAB1.0	- 25	- 12	- B	- 8	- 8	KC120
GEAHB1.0	- 40	- 6	- A	- 10	- 10	KTC20-K4.0

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore
Code	KC90, KC120	TPC	DHL, DHR	WDH
Spec.	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. Not applicable to Shape A. Not applicable to Straight Bore Type.	Changes the tapped hole dimension. Ordering Code TPC4 Not applicable to Shape A. Not applicable to Straight Bore Type. L1, L2 > TPC/2	Changes shaft bores to stepped bores. Not applicable to Shape K. (Z: 1mm Increment, J: 0.1mm Increment) Ordering Code DHL-Z20-K4.0. Applicable to Straight Bore Type Only.	Changes shaft bores to both ends stepped hole. Not applicable to Shape K. (Q, R, S, T: 1mm Increment) Ordering Code WDH-Q10-R10-S3-T3. Applicable to Straight Bore Type Only.

Spur Gear

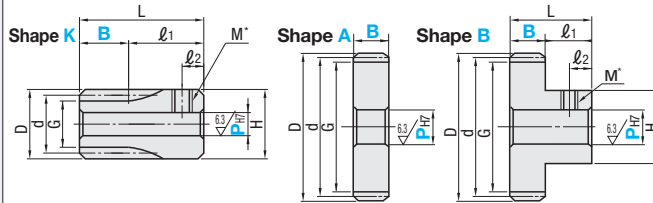
Pressure Angle 20°, Module 1.5, Shaft Bore Configurable Type



Type			M Material	S Surface Treatment	A Accessory
Straight Bore	Straight Bore + Tap	Keyway, Keyway + Tap			
GEAHB	GEAB	GEAKB	S45C Equivalent	-	Set Screw (SCM435, Black Oxide)
GEAHBB	GEABB	GEAKBB		Black Oxide	
GEAHBG	GEABG	GEAKBG		Electroless Nickel Plating	
GEAHS	GEAS	GEAKS	SUS304	-	Set Screw (SUS304)

- Set Screw is not included in Un-tapped Type products.
- Selectable Gear Shapes differ depending on the number of teeth. Check the spec. table.

Gear Shape



Tapped shaft bores are not available for Shape A.

Shaft Bore Specifications (Selectable Gear Shapes)	
Straight Bore (Shape A, Shape B, Shape K)	Keyway + Tap (Shape B, Shape K)

- For Tap, Keyway Dimension Details, see P.1532.
- Positioning of keyway and teeth is not fixed.

Accuracy Previous JIS B 1702 Class 4 (New JIS B 1702-1 Class 8 Equivalent)

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on P.1557. For configuring the tooth width / hub dimensions, see P.1551.

Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. P.H7 (1mm Increment)		d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ1	ℓ2	M (Coarse)	*1 Allowable Transmission Force (N·m) Bending Strength	
Type	Module				Straight Bore Straight Bore + Tap	Keyway, Keyway + Tap									S45C Equivalent	SUS304
Straight Bore (Shape A, Shape B, Shape K) GEAHB GEAHBB GEAHBG GEAHS	1.5	12	15	A K	6~12	8N	18	21	14.25	21	30	15	4	M4	8.33	4.76
		13			6~13		19.5	22.5	15.75	22.5					9.55	5.45
		14			6~8		21	24	17.25	16					10.79	6.16
		*15					22.5	25.5	18.75	18					12.09	-
		16			6~12		24	27	20.25	20					13.39	7.64
		17					25.5	28.5	21.75	21					14.72	8.4
		18			6~13	8N~12N	27	30	23.25	22	16.06	9.17				
		19			6~15		28.5	31.5	24.75	24	17.41	9.94				
		20			6~16		30	33	26.25	25	18.81	10.74				
		21			6~17		31.5	34.5	27.75		20.2	11.52				
		22			6~18	33	36	29.25	26	21.63	12.34					
		23			6~18	34.5	37.5	30.75	27	23.02	13.14					
		24			6~20	36	39	32.25	30	24.45	13.95					
		25			6~22	37.5	40.5	33.75		25.92	14.79					
		26			6~22	39	42	35.25	32	27.42	15.65					
		27			6~23	40.5	43.5	36.75	34	28.85	16.46					
		28				42	45	38.25		30.43	17.36					
		29			6~24	43.5	46.5	39.75	35	31.79	18.14					
		30				45	48	41.25		33.3	19					
		Straight Bore + Tap (Shape B, Shape K) GEAB GEABB GEABG GEAS			1.5	*32	A B	6~28	8N~24N	48	51	44.25	40	27	12	5
34	51		54	47.25		39.45				22.51						
35	52.5		55.5	48.75		40.93				23.35						
36	54		57	50.25		42.24				24.1						
38	57		60	53.25		45.44				25.93						
40	60		63	56.25		48.31				27.57						
42	8~33		10N~30N	63		66		59.25	51.43	29.35						
44				66		69		62.25	54.62	31.17						
45				67.5		70.5		63.75	56.23	32.09						
46				69		72		65.25	57.86	33.02						
48	8~38	10N~33N	72	75	68.25	60.9	34.75									
50			75	78	71.25	63.97	36.51									
52			78	81	74.25	67.07	38.28									
54			81	84	77.25	70.21	40.07									
Keyway (Shape A) Keyway + Tap (Shape B, Shape K) GEAKB GEAKBB GEAKBG GEAKS	1.5	55	A B	12~38	12N~33N	82.5	85.5	78.75	50	27	12	5	M6	71.94	41.06	
		56				84	87	80.25						73.7	42.05	
		58				87	90	83.25						76.57	43.7	
		60				90	93	86.25						79.8	45.54	
		62				93	96	89.25						82.69	47.19	
		64				96	99	92.25						85.97	49.06	
		65		12~43	12N~37N	97.5	100.5	93.75	87.43	49.89						
		66				99	102	95.25	89.28	50.95						
		68				102	105	98.25	92.62	52.85						
		70				105	108	101.25	95.56	54.53						
72	12~50	12N~44N	108	111	104.25	98.5	56.21									
75			112.5	115.5	108.75	103.39	59									
*80			120	123	116.25	111.3	-									
84			126	129	122.25	117.79	67.22									
85			127.5	130.5	123.75	119.29	68.07									
90			135	138	131.25	127.36	72.68									
95			142.5	145.5	138.75	135.52	77.33									
100			150	153	146.25	143.75	82.03									
120	12~54	12N~47N	180	183	176.25	70	175.06	99.9								

Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P.1-837	P.1-901	P.1-1521	P.2-277	P.1-947	P.1-1017	P.1-297
P.1-900	P.1-926	P.1-1530	P.2-282	P.1-1014	P.1-1049	P.1-330

⚠ For products uncovered by the e-Catalog Standard, see P.131.

Number of Teeth	Unit Price									
	Straight Bore				Straight Bore + Tap				Keyway, Keyway + Tap	
	GEAHB	GEAHBB	GEAHBG	GEAHS	GEAB	GEABB	GEABG	GEAS	GEAKB	GEAKBB
12										
13										
14										
15										
16										
17										
18										
19										
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85										
90										
95										
100										
120										

Alterations	Part Number	Number of Teeth	B	Order Shape	P	(KC90, KC120, TPC, DHL, DHR, WDH, LFC, KFC, KTC, QFC, QTC)
	GEAKBG1.5	30	15	B	10N	KC120
	GEAHBG1.5	60	15	A	30	KTC60-K6.0

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore							
Code	KC90, KC120	TPC	DHL, DHR	WDH							
Spec.	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type.	Changes the tapped hole dimension. [Ordering Code] TPC4 ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type.	Changes shaft bores to stepped bores. (Z: 1mm Increment, J: 0.1mm Increment) [Ordering Code] DHL-Z20-J4.0 ⊗ Applicable to Straight Bore Type Only.	Changes shaft bores to both ends stepped hole. (Q, R, S, T: 1mm Increment) [Ordering Code] WDH-Q10-R10-S5-T5 ⊗ Applicable to Straight Bore Type Only.							
	 	<table border="1"><tr><td>M</td><td>TPC</td></tr><tr><td>M4</td><td>M3 M5</td></tr><tr><td>M5</td><td>M4 M6</td></tr><tr><td>M6</td><td>M5 M8</td></tr></table>	M	TPC	M4	M3 M5	M5	M4 M6	M6	M5 M8	 • DHL: $P+2 \leq Z \leq G-4, 2 \leq J \leq 12$ ⊗ Shape B: $P+2 \leq Z \leq H-4, 2 \leq J \leq 1$ • DHR: $P+2 \leq Z \leq G-4, 2 \leq J \leq L-3$
M	TPC										
M4	M3 M5										
M5	M4 M6										
M6	M5 M8										

Alterations	Side Counterbored	Side Slotted Hole	Side Through Hole	Side Tapped Hole
Code	ZFC, ZTC	LFC, LTC	KFC, KTC	QFC, QTC
Spec.	Machines counterbored holes on the side surface. (ZFC, ZTC: 1mm Increment) ⊗ P+U+3≤ZFC(ZTC)≤G-U-4 ⊗ Applicable to Shape A only. [J Selection] U3, U4, U5, U6 [Ordering Code] ZFC20-U3	Machines slotted holes on the side surface (30°). (LFC, LTC: 1mm Increment) ⊗ P+C+4≤LFC(LTC)≤G-C-4 ⊗ Applicable to Shape A only. [M Selection] M3 M4 M5 M6 [Ordering Code] LFC20-M3	Machines through holes on the side surface. (KFC, KTC: 1mm Increment) K: 0.5mm Increment ⊗ Applicable to Shape A only. ⊗ P+K+4≤KFC(KTC)≤G-K-4 [K Selection] K3.0-K6.0 [Ordering Code] KFC20-K3.5	Machines tapped holes on the side surface of the gear. (QFC, QTC: 1mm Increment) ⊗ Applicable to Shape A only. ⊗ P+M+4≤QFC(QTC)≤G-M-4 [M Selection] M3, M4 [Ordering Code] QFC25-M3 ⊗ Tapped Hole Depth Mx1.5 (When B<Mx1.5, through)

Spur Gear

Pressure Angle 20°, Module 2.0, Shaft Bore Configurable Type



RoHS10

Type			Material	Surface Treatment	Accessory
Straight Bore	Straight Bore + Tap	Keyway, Keyway + Tap			
GEAHB	GEAB	GEAKB	S45C Equivalent	-	Set Screw (SCM435, Black Oxide)
GEAHBB	GEABB	GEAKBB		Black Oxide	
-	GEABG	GEAKBG		Electroless Nickel Plating	Set Screw (SUS304)
GEAHS	-	GEAKS	SUS304	-	

⚠ Set Screw is not included in Un-tapped Type products.

Gear Shape

Shape A

Shape B

Shaft Bore Specifications (Selectable Gear Shapes)

Straight Bore (Shape A, Shape B)	Straight Bore + Tap (Shape B)
Keyway (Shape A)	Keyway + Tap (Shape B)

⚠ For Tap, Keyway Dimension Details, see **P.1532**.
⚠ Positioning of keyway and teeth is not fixed.

Accuracy Previous JIS B 1702 Class 4
(New JIS B 1702-1 Class 8 Equivalent)

⚠ Tapped shaft bores are not available for Shape A.

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on **P.1557**. For configuring the tooth width / hub dimensions, see **P.1551**.

Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. P _{H7} (1mm Increment)		d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ ₁	ℓ ₂	M (Coarse)	*1 Allowable Transmission Force (N·m) Bending Strength	
Type	Module				Straight Bore Straight Bore + Tap	Keyway Keyway + Tap									S45C Equivalent	SUS304
Straight Bore (Shape A, Shape B) GEAHB GEAHBB GEAHS	2	12	20	A	8		24	28	19	18	30	10	5	M5	19.75	11.27
		13			8~11	8N	26	30	21	20					22.65	12.92
		14					28	32	23						25.58	14.6
		15			8~15		30	34	25	24					28.65	16.35
		16			8~16	8N~12N	32	36	27	25					31.73	18.11
		17			8~19	8N~14N	34	38	29	28					34.89	19.91
		18				8N~16N	36	40	31	30					38.07	21.73
		19			8~21	8N~17N	38	42	33	31					41.27	23.55
		20				8N~18N	40	44	35	33					44.59	25.45
		21			8~23	8N~19N	42	46	37	34					47.87	27.32
		22				8N~21N	44	48	39	36					51.27	29.26
		23			8~25	8N~22N	46	50	41	37					54.57	31.14
		24					48	52	43						57.96	33.07
		25			8~28	8N~23N	50	54	45	40					61.43	35.05
		26			8~30	8N~27N	52	56	47	42					64.99	37.09
		27					54	58	49						68.39	39.02
		28			8~33	8N~30N	56	60	51	45					72.12	41.16
		29			8~35	8N~31N	58	62	53	47					75.35	43
		30			8~36	8N~32N	60	64	55	48					78.93	45.04
		32			8~38	8N~33N	64	68	59						85.59	48.84
		34			10~38	10N~33N	68	72	63	50	93.52	53.37				
		35			10~40	10N~35N	70	74	65	52	97.01	55.36				
		36					72	76	67		100.13	57.14				
		38			10~43	10N~37N	76	80	71	55	107.7	61.46				
		40					80	84	75		114.52	65.35				
		42					84	88	79		121.92	69.57				
		44			10~48	10N~40N	88	92	83	60	129.46	73.88				
		45					90	94	85		133.29	76.06				
46						92	96	87		137.15	78.26					
48						96	100	91		144.35	82.38					
50		10~50			10N~42N	100	104	95	62	151.64	86.53					
52						104	108	99		158.99	90.73					
54						108	112	103		166.42	94.97					
55		12~52			12N~44N	110	114	105		170.54	97.32					
56						112	116	107		174.69	99.69					
58						116	120	111		181.5	103.58					
60						120	124	115		189.16	107.94					
62						124	128	119		196.02	111.86					
64		15~52			15N~44N	128	132	123		203.78	116.29					
65						130	134	125		207.23	118.26					
68						136	140	131		219.54	125.28					
70						140	144	135		226.51	129.26					
*72						144	148	139		233.49						
*75		20~52			20N~44N	150	154	145		245.07						
*80						160	164	155		263.82						
*85						170	174	165		282.76						
*90		20~55			20N~47N	180	184	175		301.9						
*95						190	194	185		321.22						
*100						200	204	195		262.11						

⚠ * marked number of teeth is not available for GEAKS, GEAKS. ⚠ The "-" text on the above table means that any Shaft Bore Dia. is not selectable.

⚠ Shaft bore diameter 9N is not available for Keyway Bore + Tap.

⚠ Specify 10K as the P dimension if keyway width of 4.0mm (height 1.8mm) for Keyway + Tap with shaft bore diameter of 10 is desired **P.1532**.

*1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions. For conditions, see **P.1534**.



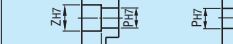
Part Number	Number of Teeth	B	Gear Shape	P
GEAKBG2.0	15	20	B	10N
GEAHBB2.0	30	20	A	8

Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837	P1-901	P1-1521	P2-277	P1-947	P1-1017	P1-297
P1-900	P1-926	P1-1530	P2-282	P1-1014	P1-1049	P1-330

☞ For products uncovered by the e-Catalog Standard, see ☞ P.131.

Number of Teeth	Unit Price									
	Straight Bore			Straight Bore + Tap			Keyway, Keyway + Tap			
	GEAHB	GEAHBB	GEAHS	GEAB	GEABB	GEABG	GEAKB	GEAKBB	GEAKBG	GEAKS
12										
13										
14										
15										
16										
17										
18										
19										
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72										
75										
80										
85										
90										
95										
100										

Alterations	Part Number	Order Code	Order Code	Order Code	Order Code	Order Code	Order Code	Order Code	Order Code	Order Code
	GEAB2.0	15	20	B	8	KC120				
	GEAHS2.0	50	20	A	20	WDH-Q25-R25-S6-T5				

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore						
Code	KC90, KC120	TPC	DHL, DHR	WDH						
Spec.	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. ❌ Not applicable to Shape A ❌ Not applicable to Straight Bore Type. 	Changes the tapped hole dimension. Ordering Code TPC4 ❌ Not applicable to Shape A ❌ Not applicable to Straight Bore Type. <table><tr><td>M</td><td>TPC</td></tr><tr><td>M5</td><td>M4 M6</td></tr><tr><td>M6</td><td>M5 M8</td></tr></table>	M	TPC	M5	M4 M6	M6	M5 M8	Changes shaft bores to stepped bores. (Z: 1mm Increment, J: 0.1mm Increment) Ordering Code DHL-Z20-J4.0 ❏ Applicable to Straight Bore Type Only.  • DHL: $J \pm 0.1$ • DHR: $J \pm 0.1$ ❏ Shape A: $P+2 \leq Z \leq G-4, 2 \leq J \leq 17$ ❏ Shape B: $P+2 \leq Z \leq G-4, 2 \leq J \leq 13$	Changes shaft bores to both ends stepped hole. (Q, R, S, T: 1mm Increment) ❏ S, T ≥ 3 Ordering Code WDH-Q10-R10-S5-T5 ❏ Applicable to Straight Bore Type Only.  • Shape A: $P+2 \leq Q, R \leq G-4$ ❏ S+T ≤ 22 ❏ S+T ≤ L-3 ❏ Shaft Bore Dia. P is general tolerance. • Shape B: $P+2 \leq Q, R \leq H-4$
	M	TPC								
M5	M4 M6									
M6	M5 M8									

Alterations	Side Counterbored	Side Slotted Hole	Side Through Hole	Side Tapped Hole
Code	ZFC, ZTC	LFC, LTC	KTC, KFC	QFC, QTC
Spec.	Machines counterbored holes on the side surface. (ZFC, ZTC: 1mm Increment) ☞ P+U1+3≤ZFC(ZTC)≤G-U1-4 ☞ Applicable to Shape A only. U Selection U3, U4, U5, U6 Ordering Code ZFC20-U3	Machines slotted holes on the side surface (30°). (LFC, LTC: 1mm Increment) ☞ P+C+4≤LFC(LTC)≤G-C-4 ☞ Applicable to Shape A only. M Selection M3 M4 M5 M6 Ordering Code LFC20-M3	Machines through holes on the side surface. (KFC, KTC: 1mm Increment) K: 0.5mm Increment ☞ Applicable to Shape A only. ☞ P+K+4≤KFC(KTC)≤G-K-4 K Selection K3.0-K6.0 Ordering Code KFC20-K3.5	Machines tapped holes on the side surface of the gear. (QFC, QTC: 1mm Increment) ☞ Applicable to Shape A only. ☞ P+M+4≤QFC(QTC)≤G-M-4 M Selection M3, M4 Ordering Code QFC25-M3 ☞ Tapped Hole Depth Mx1.5 (When B<Mx1.5, through)

Spur Gear

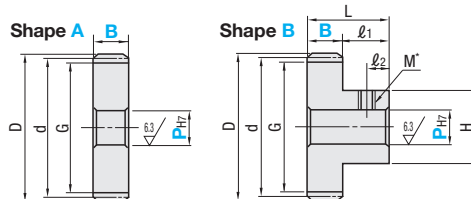
Pressure Angle 20°, Module 2.5, Shaft Bore Configurable Type



Type			Material	Surface Treatment	Accessory
Straight Bore	Straight Bore + Tap	Keyway, Keyway + Tap			
GEAHB	GEAB	GEAKB	S45C	-	Set Screw (SCM435, Black Oxide)
GEAHBB	GEABBB	GEAKBB	Equivalent	Black Oxide	

⚠ Set Screw is not included in Un-tapped Type products.

Gear Shape



⊗ Tapped shaft bores are not available for Shape A.

Shaft Bore Specifications (Selectable Gear Shapes)	
Straight Bore (Shape A, Shape B)	Straight Bore + Tap (Shape B)
Keyway (Shape A)	Keyway + Tap (Shape B)

⚠ For Tap, Keyway Dimension Details, see **P.1532**
⚠ Positioning of keyway and teeth is not fixed.

Accuracy Previous JIS B 1702 Class 4
(New JIS B 1702-1 Class 8 Equivalent)

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on **P.1557**. For configuring the tooth width / hub dimensions, see **P.1551**.

Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. Ph7 (1mm Increment)		d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ1	ℓ2	M (Coarse)	*1 Allowable Transmission Force (N·m) Bending Strength	
Type	Module				Straight Bore	Keyway, Keyway + Tap									S45C Equivalent	
Straight Bore (Shape A, Shape B) GEAHB GEAHBB	2.5	12	25	A B	8~13	8N	30	35	23.75	23	37	12	6	M5	38.58	
		14			8~15	8N~12N	35	40	28.75	25					49.96	
		15			8~20	8N~15N	37.5	42.5	31.25	30					55.96	
		16			8~22	8N~17N	40	45	33.75	32					61.98	
		17			8~24	8N~19N	42.5	47.5	36.25	35					68.13	
		18			8~26	8N~22N	45	50	38.75	38					74.36	
		19			8~27		47.5	52.5	41.25	39					80.6	
		20			8~28	8N~23N	50	55	43.75	40					87.09	
		21			8~30	8N~27N	52.5	57.5	46.25	42					93.5	
		22			8~32	8N~29N	55	60	48.75	44					100.13	
		23			8~34	8N~30N	57.5	62.5	51.25	46					106.58	
		24			8~36	8N~32N	60	65	53.75	48					113.19	
		25			8~38	8N~33N	62.5	67.5	56.25	50					119.98	
		26			8~43	8N~37N	65	70	58.75	55					126.94	
		27			8~48	8N~40N	67.5	72.5	61.25	60					133.57	
		70					75	63.75	140.86							
		Straight Bore + Tap (Shape B) GEAB GEABBB			29	8~49	8N~41N	72.5	77.5	66.25					62	147.18
					30	8~51	8N~43N	75	80	68.75					65	154.16
32					8~55	8N~47N	80	85	73.75	70				167.17		
34					10~55	10N~47N	85	90	78.75					182.65		
35							87.5	92.5	81.25					189.48		
36							90	95	83.75	195.56						
38					10~57	10N~49N	95	100	88.75	75				210.35		
40							100	105	93.75	80				223.66		
42							105	110	98.75					238.12		
44					10~60	10N~50N	110	115	103.75					80	252.85	
45							112.5	117.5	106.25	260.32						
46							115	120	108.75	267.86						
48					10~60	12N~50N	120	125	113.75	85				281.94		
50							125	130	118.75					296.17		
52							130	135	123.75					310.53		
54					15~60	15N~50N	135	140	128.75	325.04						
55							137.5	142.5	131.25	333.08						
56							140	145	133.75	341.19						
58		145					150	138.75	354.5							
60		150					155	143.75	369.44							
62		155					160	148.75	382.84							
64		25~60			15N~50N	160	165	153.75	398.01							
65						162.5	167.5	156.25	404.75							
68						170	175	163.75	428.78							
70						175	180	168.75	442.41							
72						180	185	173.75	456.04							
75						187.5	192.5	181.25	478.65							
80						200	205	193.75	396.36							

⊗ Shaft bore diameter 9N is not available for Keyway Bore + Tap. ⚠ The "-" text on the above table means that any Shaft Bore Dia. is not selectable.

⚠ Specify 10K as the P dimension if keyway width of 4.0mm (height 1.8mm) for Keyway + Tap with shaft bore diameter of 10 is desired **P.1532**.

*1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions. For conditions, see **P.1534**.

Ordering Example	Part Number	Number of Teeth	B	Gears Shape	P
	GEAB2.5	15	25	B	8
	GEAKB2.5	30	25	A	10N

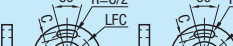
Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837	P1-901	P1-1521	P2-277	P1-947	P1-1017	P1-297
P1-900	P1-926	P1-1530	P2-282	P1-1014	P1-1049	P1-330

☞ For products uncovered by the e-Catalog Standard, see ☞ P.131.

Number of Teeth	Unit Price					
	Straight Bore		Straight Bore + Tap		Keyway, Keyway + Tap	
	GEAHB	GEAHBB	GEAB	GEABB	GEAKB	GEAKBB
12						
14						
15						
16						
17						
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80						

Alterations	Part Number	Number of Teeth	B	Gear Shape	P	(KC90, KC120, TPC, DHL, DHR, WDH, LFC, LTC, KFC, KTC, QFC, QTC)
	GEAB2.5	15	25	B	8	TPC4
	GEAHBB2.5	30	25	A	15	DHL-Z30-J10

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore					
Code	KC90, KC120	TPC	DHL, DHR	WDH					
Spec.	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type.	Changes the tapped hole dimension. Ordering Code TPC4 ⊗ Not applicable to Shape A. ⊗ Not applicable to Straight Bore Type.	Changes shaft bores to stepped bores. (Z: 1mm Increment, J: 0.1mm Increment) Ordering Code DHL-Z20-J4.0 ♡ Applicable to Straight Bore Type Only.	Changes shaft bores to both ends stepped hole. (Q, R, S, T: 1mm Increment) ♡ S, T ≥ 3 Ordering Code WDH-Q10-R10-S5-T5 ♡ Applicable to Straight Bore Type Only.					
	 	<table><tr><td>M</td><td>TPC</td></tr><tr><td>M5</td><td>M4 M6</td></tr><tr><td>M6</td><td>M5 M8</td></tr></table>	M	TPC	M5	M4 M6	M6	M5 M8	 
M	TPC								
M5	M4 M6								
M6	M5 M8								

Alterations Code	Side Slotted Hole LFC, LTC	Side Through Hole KFC, KTC	Side Tapped Hole QFC, QTC										
	Machines slotted holes on the side surface (30°). (LFC, LTC: 1mm Increment) P+C+4≤LFC(LTC)≤G-C-4 Applicable to Shape A only. M Selection M3 M4 M5 M6 Ordering Code LFC20-M3	Machines through holes on the side surface. (KFC, KTC: 1mm Increment, K: 0.5mm Increment) P+K+4≤KFC(KTC)≤G-K-4 K Selection K3.0~K6.0 Ordering Code KFC20-K3.5	Machines tapped holes on the side surface of the gear (QFC, QTC: 1mm Increment). Applicable to Shape A only. P+M+4≤QFC(QTC)≤G-M-4 M Selection M3, M4 Ordering Code QFC25-M3 Tapped Hole Depth Mx1.5 (When B<Mx1.5, through)										
Spec.	<table><tr><td>M</td><td>C</td></tr><tr><td>M3</td><td>3.5</td></tr><tr><td>M4</td><td>4.5</td></tr><tr><td>M5</td><td>5.5</td></tr><tr><td>M6</td><td>6.5</td></tr></table> 	M	C	M3	3.5	M4	4.5	M5	5.5	M6	6.5		
M	C												
M3	3.5												
M4	4.5												
M5	5.5												
M6	6.5												

Spur Gear

Pressure Angle 20°, Module 3.0, Shaft Bore Configurable Type

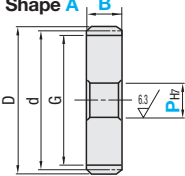


Type		Material	Surface Treatment	Accessory
Straight Bore	Keyway, Keyway + Tap			
GEAHB	GEAKB	S45C Equivalent	-	Set Screw (SCM435, Black Oxide)
GEAHBB	GEAKBB		Black Oxide	
GEAHBG	GEAKBG		Electroless Nickel Plating	
GEAHS	GEAKS	SUS304	-	Set Screw (SUS304)

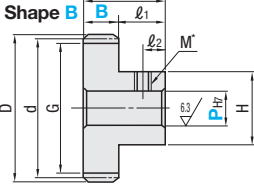
⚠ Set Screw is not included in Un-tapped Type products.

Gear Shape

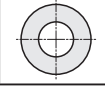
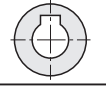
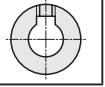
Shape A B



Shape B



Shaft Bore Specifications (Selectable Gear Shapes)

Straight Bore (Shape A, Shape B)	Keyway (Shape A)	Keyway + Tap (Shape B)
		

⚠ For Tap, Keyway Dimension Details, see **P1532**.
 ⚠ Positioning of keyway and teeth is not fixed.

RoHS10

⚠ Tapped shaft bores are not available for Shape A.

Accuracy Previous JIS B 1702 Class 4
(New JIS B 1702-1 Class 8 Equivalent)

When desiring to fix the gear hub onto the shaft through MechaLock, see Keyless Type on **P1557**. For configuring the tooth width / hub dimensions, see **P1551**.

Part Number		Number of Teeth	B	Gear Shape	Shaft Bore Dia. PH7 (1mm Increment)		d Reference Dia.	D Tip Dia.	G Root Dia.	H	L	ℓ ₁	ℓ ₂	M (Coarse)	*1 Allowable Transmission Force (N・m) Bending Strength	
Type	Module				Straight Bore	Keyway Keyway + Tap									S45C Equivalent	SUS304
Straight Bore (Shape A, Shape B) GEAHB GEAHBB GEAHBG	3	12	30	A	8~17	8N~12N	36	42	28.5	27	48	18	9	M5	66.66	38.04
		14			8~22	8N~16N	42	48	34.5	32					86.33	49.27
		15			8~24	8N~19N	45	51	37.5	35					96.7	55.18
		16			8~28	8N~23N	48	54	40.5	40					107.1	61.12
		51					57	43.5	117.74	67.19						
		17			8~32	8N~27N	54	60	46.5	44					128.5	73.33
		18			8~33	8N~28N	57	63	49.5	45					139.28	79.48
		19			8~38	8N~33N	60	66	52.5	50					150.49	85.88
		20			8~40	8N~35N	63	69	55.5	52					161.57	92.2
		21			8~42	8N~36N	66	72	58.5	54					173.02	98.74
		22			8~44	8N~37N	69	75	61.5	56					184.17	105.09
		23			8~46	8N~39N	72	78	64.5	58					195.6	111.62
		24			8~48	8N~40N	75	81	67.5	60					207.33	118.31
		25			10~53	10N~44N	78	84	70.5	65					219.36	125.18
		81					87	73.5	230.8	131.71						
		26			10~58	10N~47N	84	90	76.5	70					243.41	138.9
		87					93	79.5	254.32	145.13						
		27			10~60	10N~50N	90	96	82.5	75				266.4	152.02	
		96					102	88.5	288.88	164.85						
		28			12~63	12N~50N	102	108	94.5	80				315.62	180.11	
		105					111	97.5	327.41					186.84		
		108					114	100.5	337.93					192.84		
		114					120	106.5	363.48					207.42		
		29			12~65		120	126	112.5					386.49	220.55	
		126					132	118.5	411.47					234.81		
		30			20~65	20N~50N	132	138	124.5					436.93	249.34	
		135					141	127.5	449.84					256.7		
		32			25~68	25N~50N	138	144	130.5	85				462.87	264.14	
		144					150	136.5	487.2					278.02		
		150					156	142.5	511.77					292.04		
		156					162	148.5	536.6					306.21		
		162					168	154.5	561.67					320.52		
		165					171	157.5	575.63					328.44		
		168					174	160.5	589.57					336.44		
		174					180	166.5	612.58					349.57		
		180					186	172.5	638.4					364.31		
		186					192	178.5	661.55							
		192					198	184.5	529.05							
		195					201	187.5	538					-		
		34			25~70		204	210	196.5	90				569.95		
		210					216	202.5	588.06							
35	12~63	12N~50N		105			111	97.5	80		327.41	186.84				
108				114			100.5	337.93			192.84					
114				120			106.5	363.48			207.42					
120				126			112.5	386.49			220.55					
126				132	118.5	411.47	234.81									
132				138	124.5	436.93	249.34									
36	25~68	25N~50N		135	141	127.5	85	449.84	256.7							
138				144	130.5	462.87		264.14								
144				150	136.5	487.2		278.02								
150				156	142.5	511.77		292.04								
156				162	148.5	536.6		306.21								
162				168	154.5	561.67		320.52								
165			171	157.5	575.63	328.44										
168			174	160.5	589.57	336.44										
174			180	166.5	612.58	349.57										
180			186	172.5	638.4	364.31										
186			192	178.5	661.55											
38			25~70		192	198		184.5	90	529.05						
195	201	187.5			538	-										
204	210	196.5			569.95											
210	216	202.5			588.06											
39	12~65				120	126	112.5			386.49	220.55					
126					132	118.5	411.47			234.81						
132			138	124.5	436.93	249.34										
135			141	127.5	449.84	256.7										
138			144	130.5	462.87	264.14										
144			150	136.5	487.2	278.02										
40	25~68	25N~50N	150	156	142.5	85	511.77	292.04								
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
192			198	184.5	529.05											
195			201	187.5	538		-									
42			25~70		204		210	196.5	90	569.95						
210					216		202.5	588.06								
43	12~63	12N~50N			105	111	97.5	80		327.41	186.84					
108					114	100.5	337.93			192.84						
114					120	106.5	363.48			207.42						
120					126	112.5	386.49			220.55						
126			132	118.5	411.47	234.81										
132			138	124.5	436.93	249.34										
44	25~68	25N~50N	135	141	127.5	85	449.84	256.7								
138			144	130.5	462.87		264.14									
144			150	136.5	487.2		278.02									
150			156	142.5	511.77		292.04									
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
45			25~70		192		198	184.5	90	529.05						
195	201	187.5			538	-										
204	210	196.5			569.95											
210	216	202.5			588.06											
46	12~65				120	126	112.5			386.49	220.55					
126					132	118.5	411.47			234.81						
132			138	124.5	436.93	249.34										
135			141	127.5	449.84	256.7										
138			144	130.5	462.87	264.14										
144			150	136.5	487.2	278.02										
48	25~68	25N~50N	150	156	142.5	85	511.77	292.04								
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
192			198	184.5	529.05											
195			201	187.5	538		-									
50			25~70		204		210	196.5	90	569.95						
210					216		202.5	588.06								
51	12~63	12N~50N			105	111	97.5	80		327.41	186.84					
108					114	100.5	337.93			192.84						
114					120	106.5	363.48			207.42						
120					126	112.5	386.49			220.55						
126			132	118.5	411.47	234.81										
132			138	124.5	436.93	249.34										
52	25~68	25N~50N	135	141	127.5	85	449.84	256.7								
138			144	130.5	462.87		264.14									
144			150	136.5	487.2		278.02									
150			156	142.5	511.77		292.04									
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
54			25~70		192		198	184.5	90	529.05						
195	201	187.5			538	-										
204	210	196.5			569.95											
210	216	202.5			588.06											
55	12~65				120	126	112.5			386.49	220.55					
126					132	118.5	411.47			234.81						
132			138	124.5	436.93	249.34										
135			141	127.5	449.84	256.7										
138			144	130.5	462.87	264.14										
144			150	136.5	487.2	278.02										
56	25~68	25N~50N	150	156	142.5	85	511.77	292.04								
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
192			198	184.5	529.05											
195			201	187.5	538		-									
58			25~70		204		210	196.5	90	569.95						
210					216		202.5	588.06								
59	12~63	12N~50N			105	111	97.5	80		327.41	186.84					
108					114	100.5	337.93			192.84						
114					120	106.5	363.48			207.42						
120					126	112.5	386.49			220.55						
126			132	118.5	411.47	234.81										
132			138	124.5	436.93	249.34										
60	25~68	25N~50N	135	141	127.5	85	449.84	256.7								
138			144	130.5	462.87		264.14									
144			150	136.5	487.2		278.02									
150			156	142.5	511.77		292.04									
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
62			25~70		192		198	184.5	90	529.05						
195	201	187.5			538	-										
204	210	196.5			569.95											
210	216	202.5			588.06											
63	12~65				120	126	112.5			386.49	220.55					
126					132	118.5	411.47			234.81						
132			138	124.5	436.93	249.34										
135			141	127.5	449.84	256.7										
138			144	130.5	462.87	264.14										
144			150	136.5	487.2	278.02										
64	25~68	25N~50N	150	156	142.5	85	511.77	292.04								
156			162	148.5	536.6		306.21									
162			168	154.5	561.67		320.52									
165			171	157.5	575.63		328.44									
168			174	160.5	589.57		336.44									
174			180	166.5	612.58		349.57									
180			186	172.5	638.4		364.31									
186			192	178.5	661.55											
192			198	184.5	529.05											
195			201	187.5	538		-									
66			25~70		204		210	196.5	90	569.95						
210					216		202.5	588.06								
67	12~63	12N~50N			105	111	97.5	80		327.41	186.84					
108					114	100.5	337.93			192.84						
114					120	106.5	363.48			207.42						
120					126	112.5	386.49			220.55						
126			132	118.5	411.47	234.81										
132			138	124.5	436.93	249.34										
68	25~68	25N~50N	135	141	127.5	85	449.84	256.7								
138			144	130.5	462.87</											

⚠ * marked number of teeth is not available for GEAHS, GEAKS. ⚠ The "-" text on the above table means that any Shaft Bore Dia. is not selectable.

⚠ Shaft bore diameter 9N is not available for Keyway Bore + Tap.

⚠ Specify 10K as the P dimension if keyway width of 4.0mm (height 1.8mm) for Keyway + Tap with shaft bore diameter of 10 is desired **P1532**.

*1. Allowable Transmission Forces in the table are reference values calculated with prescribed conditions. For conditions, see **P1534**.



Ordering Example

Part Number	-	Number of Teeth	-	B	-	Gears Shape	-	P
GEAHB3.0	-	15	-	30	-	B	-	8
GEAKBG3.0	-	30	-	30	-	A	-	10N

Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837	P1-901	P1-1521	P2-277	P1-947	P1-1017	P1-297
P1-900	P1-926	P1-1530	P2-282	P1-1014	P1-1049	P1-330

☎ For products uncovered by the e-Catalog Standard, see ☎ P131.

Number of Teeth	Unit Price							
	Straight Bore				Keyway, Keyway + Tap			
	GEAHB	GEAHBB	GEAHBG	GEAHS	GEAKB	GEAKBB	GEAKBG	GEAKS
12								
14								
15								
16								
17								
18								
19								
20								
21								
22								
23								
24								
25								
26								
27								
28								
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62								
64								
65								
68								
70								

Alterations	Part Number	Number of Teeth	B	Gear Shape	P	(KC90, KC120, TPC, DHL, DHR, WDH, LFC, LTC, KFC, KTC, QFC, QTC)
	GEAHB3.0	30	30	A	20	KFC40-K6.0

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore					
Code	KC90, KC120	TPC	DHL, DHR	WDH					
Spec.	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. ✖ Not applicable to Shape A. ✖ Not applicable to Straight Bore Type.	Changes the tapped hole dimension. [Ordering Code] TPC4 ✖ Not applicable to Shape A. ✖ Not applicable to Straight Bore Type.	Changes shaft bores to stepped bores. (Z: 1mm Increment, J: 0.1mm Increment) [Ordering Code] DHL-Z20-J4.0 ✖ Applicable to Straight Bore Type Only.	Changes shaft bores to both ends stepped bore. (Q, R, S, T: 1mm Increment) ✖ S, T≥3 [Ordering Code] WDH-Q10-R10-S5-T5 ✖ Applicable to Straight Bore Type Only.					
	 	<table><tr><td>M</td><td>TPC</td></tr><tr><td>M5</td><td>M4 M6</td></tr><tr><td>M6</td><td>M5 M8</td></tr></table>	M	TPC	M5	M4 M6	M6	M5 M8	  • DHL ✖ Shape A: P+2≤Z≤G-4, 2≤J≤27 ✖ Shape B: P+2≤Z≤H-4, 2≤J≤18 ✖ Shape B: P+2≤Z≤G-4, 2≤J≤45
M	TPC								
M5	M4 M6								
M6	M5 M8								

Alterations	Side Slotted Hole	Side Through Hole	Side Tapped Hole
Code	LFC, LTC	KFC, KTC	QFC, QTC
Spec.	Machines slotted holes on the side surface (30°). (LFC, LTC: 1mm Increment) ☎ P+C+4≤LFC(LTC)≤G-C-4 ☎ Applicable to Shape A only. [M Selection] M3 M4 M5 M6 [Ordering Code] LFC20-M3	Machines through holes on the side surface. (KFC, KTC: 1mm Increment, K: 0.5mm Increment) ☎ Applicable to Shape A only. ☎ P+K+4≤KFC(KTC)≤G-K-4 [K Selection] K3.0~K6.0 [Ordering Code] KFC20-K3.5	Machines tapped holes on the side surface of the gear (QFC, QTC: 1mm Increment). ☎ Applicable to Shape A only. ☎ P+M+4≤QFC(QTC)≤G-M-4 [M Selection] M3, M4 [Ordering Code] QFC25-M3 ☎ Tapped Hole Depth Mx1.5 (When B<Mx1.5, through)

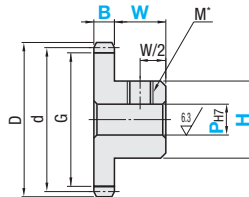
Spur Gears - Tooth Width, Hub Dimensions Configurable

Pressure Angle 20°, Module 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0



Type			Material	Surface Treatment	Accessory
Straight Bore	Straight Bore + Tap	Keyway + Tap			
GEFHB	GEFB	GEFKB	S45C Equivalent	-	Set Screw (SCM435, Black Oxide)
GEFHBB	GEFBB	GEFKBB		Black Oxide	
GEFHGB	GEFBG	GEFKBG	SUS304	Electroless Nickel Plating	Set Screw (SUS304)
GEFHBS	GEFBS	GEFKBS		-	

Set Screw is not included in Straight Bore products.



Shaft Bore Specifications		
Straight Bore	Straight Bore + Tap	Keyway + Tap

Keyway Dimension Details P.1532

Positioning of keyway and teeth is not fixed.

Tapped Hole Dimension List

Shaft Bore Dia. P.H7	M (Coarse)	Accessory: Set Screw
3-5	M3	M3x3
6-12	M4	M4x3
13-17	M5	M5x4
18-30	M6	M6x5
31-40	M8	M8x6

Accuracy Previous JIS B 1702 Class 4
(New JIS B 1702-1 Class 8 Equivalent)

Module 0.5, 0.8, 1.0, 1.5

Part Number		Number of Teeth	B 1mm Increment	Shaft Bore Dia. P.H7 (1mm Increment)		Hub 1mm Increment		d Reference Dia.	D Tip Dia.	G Root Dia.	Unit Price											
Type	Module			Straight Bore	Keyway + Tap	W	H				Straight Bore			Straight Bore + Tap			Keyway + Tap					
				GEFHB	GEFB	GEFKB					GEFHB	GEFHBB	GEFBG	GEFBS	GEFKB	GEFKBB	GEFKBG	GEFKBS				
Straight Bore	0.5	20	2-5	3-4	5	0-10	P+2≤H≤G-2 (Straight Bore)	10	11	8.75												
		24						12	13	10.75												
		25						12.5	13.5	11.25												
		26						13	14	11.75												
		30						15	16	13.75												
		32						16	17	14.75												
		36						18	19	16.75												
		40						20	21	18.75												
		48						24	25	22.75												
		50						25	26	23.75												
		60						30	31	28.75												
		70						35	36	33.75												
		80						40	41	38.75												
		100						50	51	48.75												
Straight Bore	0.8	15	3-10	5	8	0-10	P+2≤H≤G-2 (Straight Bore)	12	13.6	10												
		16						12.8	14.4	10.8												
		18						14.4	16	12.4												
		20						16	17.6	14												
		24						19.2	20.8	17.2												
		25						20	21.6	18												
		28						22.4	24	20.4												
		30						24	25.6	22												
		32						25.6	27.2	23.6												
		36						28.8	30.4	26.8												
		40						32	33.6	30												
		45						36	37.6	34												
		48						38.4	40	36.4												
		50						40	41.6	38												
Straight Bore + Tap	1.0	20	5-15	6-12	8N	0-15	P+4≤H≤G-3 (Straight Bore, Straight Bore + Tap)	20	22	17.5												
		21						21	23	18.5												
		22						22	24	19.5												
		23						23	25	20.5												
		24						24	26	21.5												
		25						25	27	22.5												
		26						26	28	23.5												
		28						28	30	25.5												
		30						30	32	27.5												
		32						32	34	29.5												
		35						35	37	32.5												
		36						36	38	33.5												
		40						40	42	37.5												
		48						48	50	45.5												
		50						50	52	47.5												
		60						60	62	57.5												
		70						70	72	67.5												
		80						80	82	77.5												
		100						100	102	97.5												
		120						120	122	117.5												
Keyway + Tap	1.5	16	10-20	6-12	8N-12N	0-20	P+6≤H≤G-4	24	27	20.25												
		18						27	30	23.25												
		20						30	33	26.25												
		24						36	39	32.25												
		25						37.5	40.5	33.75												
		26						39	42	35.25												
		28						42	45	38.25												
		30						45	48	41.25												
		35						52.5	55.5	48.75												
		36						54	57	50.25												
		40						60	63	56.25												
		48						72	75	68.25												
		50						75	78	71.25												
		60						90	93	86.25												
		70						105	108	101.25												
		80						120	123	116.25												

Fixing Parts				Others		
Rotary Shaft	Cantilever Shaft	MechaLock	Parallel Key	Bearing with Housings	Bearing	Shaft Collar
P1-837 P1-900	P1-901 P1-926	P1-1521 P1-1530	P2-277 P2-282	P1-947 P1-1014	P1-1017 P1-1049	P1-297 P1-330

☞For products uncovered by the e-Catalog Standard, see ☞ P131.

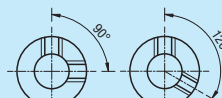
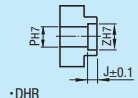
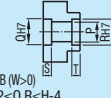
Module 2.0, 2.5, 3.0

Part Number		Number of Teeth	B 1mm Increment	Shaft Bore Dia. P _{H7} (1mm Increment)		Hub 1mm Increment		d Reference Dia.	D Tip Dia.	G Root Dia.	Unit Price											
Type	Module			Straight Bore Straight Bore + Tap	Keyway + Tap	W	H				Straight Bore				Straight Bore + Tap				Keyway + Tap			
Straight Bore GEFHB GEFHBB GEFHBG GEFHBS	2.0	15	15~25	8~15	8N~12N	0~25	P+10≤H≤G-5 (Straight Bore, Straight Bore + Tap)	30	34	25	GEFHB	GEFHBB	GEFHBBG	GEFHBS	GEFB	GEFBB	GEFBG	GEFBS	GEFKB	GEFKBB	GEFKBG	GEFKBS
		16						32	36	27												
		18						36	40	31												
		20						40	44	35												
		24						48	52	43												
		25						50	54	45												
		26						52	56	47												
		30						60	64	55												
		32						64	68	59												
		36						72	76	67												
		40						80	84	75												
		48						96	100	91												
Straight Bore + Tap GEFB GEFBB GEFBG GEFBS	2.0	50		10~30	10N~25N		P+12≤H≤G-5 (Keyway + Tap)	100	104	95												
		52		10~35	10N~30N		P+12≤H≤G-5 (Straight Bore, Straight Bore + Tap)	104	108	99												
		60		12~40	12N~35N		P+14≤H≤G-5 (Keyway + Tap)	120	124	115												
		18		8~15	8N~12N	0~30	P+10≤H≤G-6 (Straight Bore, Straight Bore + Tap)	45	50	38.75												
		20		8~20	8N~17N		P+12≤H≤G-6 (Keyway + Tap)	50	55	43.75												
		25		8~25	8N~20N		P+12≤H≤G-6 (Keyway + Tap)	62.5	67.5	56.25												
Keyway + Tap GEFKB GEFKBB GEFKBG GEFKBS	2.5	28	20~30	8~25	8N~20N		P+12≤H≤G-6 (Straight Bore, Straight Bore + Tap)	70	75	63.75												
		36		8~30	8N~25N		P+12≤H≤G-6 (Keyway + Tap)	90	95	83.75												
		40		8~30	8N~25N		P+14≤H≤G-6 (Keyway + Tap)	100	105	93.75												
		50		12~40	12N~35N		P+14≤H≤G-6 (Keyway + Tap)	125	130	118.75												
		12		8~15	8N~12N	0~35	P+10≤H≤G-7 (Straight Bore, Straight Bore + Tap)	36	42	28.5												
		15		8~20	8N~20N		P+12≤H≤G-7 (Keyway + Tap)	45	51	37.5												
		18		8~20	8N~20N		P+12≤H≤G-7 (Keyway + Tap)	54	60	46.5												
		20		8~25	8N~25N		P+12≤H≤G-7 (Keyway + Tap)	60	66	52.5												
		30		10~30	10N~30N		P+12≤H≤G-7 (Straight Bore, Straight Bore + Tap)	90	96	82.5												
		36		12~40	12N~35N		P+14≤H≤G-7 (Keyway + Tap)	108	114	100.5												
		40		12~40	12N~35N		P+14≤H≤G-7 (Keyway + Tap)	120	126	112.5												

- ☞ Shaft bore diameter 9N is not available for Keyway Bore + Tap. ☞ When specifying "Straight Bore + Tap" or "Keyway + Tap," make sure that the hub length satisfies the condition of "(W) > M+3."
- ☞ When the tooth width is smaller, the transmitting torque is smaller. ☞ Shaft bore diameter 6.35 is available for Straight Bore and Straight Bore + Tap.
- ☞ Specify 10K as the P dimension if keyway width of 4.0mm (height 1.8mm) for Keyway + Tap with shaft bore diameter of 10 is desired ☞ P1532.
- ☞ When requesting Shape A (no hub), specify W=0
- ☞ When W=0, specification for H is not required. ☞ The "-" text on the above table means that any Shaft Bore Dia. is not selectable.
- ☞ For WDH alterations, the H range requirements considered as Q and R = P must satisfy the condition of "Q + ○ ≤ H ≤ G - ○," (○: Refer to the value in the specification table)
- ☞ For DHL/DHR alterations, the H range requirements considered as Z = P must satisfy the condition of "Z + ○ ≤ H ≤ G - ○," (○: Refer to the value in the specification table)

Ordering Example	Part Number	-	Number of Teeth	-	B	-	P	-	W	-	H
	GEFHB1.0	-	30	-	8	-	15	-	W10	-	H26
	GEFKBG2.0	-	32	-	15	-	20N	-	W0	-	W0

Alterations	Part Number	-	Number of Teeth	-	B	-	P	-	W	-	H	-	(KC90, KC120...etc.)
	GEFBB1.5	-	30	-	20	-	8	-	W15	-	H30	-	KC90
	GEFHBB1.0	-	40	-	8	-	10	-	W0	-	H0	-	KTC20-K5.0

Alterations	Set Screw	Tapped Hole Dimension	Stepped Hole	Both Ends Stepped Bore												
Code	KC90, KC120	TPC	DHL, DHR	WDH												
	KC90: Adds another set screw at 90° position. KC120: Adds another set screw at 120° position. ❌ Not applicable to Straight Bore Type.	Changes the tapped hole dimension. Ordering Code: TPC4 ❌ Only when W=0 ❌ Not applicable to Straight Bore Type.	Changes shaft bores to stepped bores. (Z: 1mm Increment, J: 0.1mm Increment) Ordering Code: DHL-Z20-J4.0 ❌ Applicable to Straight Bore Type Only.	Changes shaft bores to both ends stepped hole. (Q, R, S, T: 1mm Increment) Ordering Code: WDH-Q10-R10-S5-T5 ❌ Applicable to Straight Bore Type Only.												
Spec.		<table><tr><td>M</td><td>TPC</td></tr><tr><td>M3</td><td>M4</td></tr><tr><td>M4</td><td>M3 M5</td></tr><tr><td>M5</td><td>M4 M6</td></tr><tr><td>M6</td><td>M5 M8</td></tr><tr><td>M8</td><td>M6</td></tr></table>	M	TPC	M3	M4	M4	M3 M5	M5	M4 M6	M6	M5 M8	M8	M6	 • DHL ❌ $P+2 \leq Z \leq G-4$ ❌ $2 \leq J \leq B+W-3$ • DHR ❌ $P+2 \leq Z \leq H-4$ ❌ $2 \leq J \leq W$	 • Shape A (W=0) ❌ $P+2 \leq Q, R \leq G-4$ ❌ $S+T \leq B-3$ ❌ Shaft Bore Dia. P is general tolerance. • Shape B (W>0) ❌ $P+2 \leq Q, R \leq H-4$ ❌ $S+T \leq B+W-3$
M	TPC															
M3	M4															
M4	M3 M5															
M5	M4 M6															
M6	M5 M8															
M8	M6															

Alterations	Side Counterbored ZFC, ZTC	Side Slotted Hole LFC, LTC	Side Through Hole KFC, KTC	Side Tapped Hole QFC, QTC
Code	Machines counterbored holes on the side surface. (ZFC, ZTC: 1mm Increment) ☞ P+U+3≤ZFC(ZTC)≤G-U-4 ☞ Applicable only when W=0. U Selection U1, U2, U3, U4, U5, U6 Ordering Code: ZFC20-U3	Machines slotted holes on the side surface (30°). (LFC, LTC: 1mm Increment) ☞ P+C+4≤LFC(LTC)≤G-C-4 ☞ Applicable to Shape A only. M Selection M3, M4, M5, M6 Ordering Code: LFC20-M3	Machines through holes on the side surface. (KFC, KTC: 1mm Increment) K: 0.5mm Increment ☞ Applicable to Shape A only. ☞ P+K+4≤KFC(KTC)≤G-K-4 K Selection K3.0~K6.0 Ordering Code: KFC20-K3.5	Machines tapped holes on the side surface of the gear. (QFC, QTC: 1mm Increment) ☞ Applicable to Shape A only. ☞ P+M+4≤QFC(QTC)≤G-M-4 M Selection M3, M4 Ordering Code: QFC25-M3 ☞ Tapped Hole Depth Mx1.5 (When B-Mx1.5, through)
Spec.		